Deep Patel

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PROFESSIONAL SUMMARY

High school senior focused on aerospace engineering and rocket propulsion. Creator of ORBIX, an aerospace engineering web app, and designer of a 3D-printed rocket bell nozzle studied with CFD simulation. Dedicated youth leader and volunteer looking for opportunities to learn more about physics, engineering, and space science through hands-on projects and research.

EDUCATION

Chelmsford High School, Chelmsford, MA Expected Graduation: June, 2027
GPA: 4.1 (Weighted)
Relevant Courses: AP Physics C, AP Computer Science, Calculus, Advanced Engineering Principles & 3D Design

SKILLS AND INTERESTS

Technical Skills: Java, CAD (Fusion 360, Revit, Onshape), Robotics, Microsoft Office Suite, Mathematical Modeling, Physics Problem-Solving, Adobe Creative Cloud Suite

Interests: Aerospace Engineering, Mechanical Engineering, Astrophysics, Rocketry, 3D Modeling

TECHNICAL PROJECTS

Aerodynamic Bell Nozzle Sep 2025 - Present
Lead Designer & Fabricator (Self-Directed)

- Studied nozzle thermodynamics and isentropic flow independently to size a LOX/RP-1 bell nozzle for a 100 bar chamber expanding to 0.7 bar.
- Modeled the nozzle contour in OpenSCAD and checked its 15.92 area ratio against the Mach 3.6 isentropic design point.
- Set up and debugged a compressible-flow CFD case in SimFlow, tracing solver divergence through mesh, boundary condition, and limiter changes.
- 3D-printed the 84 mm nozzle in PLA, printing it inverted so no support material formed inside the bell.

ORBIX Web Application Aug 2026 - Present
Creator & Project Lead | https://orbix-aerospace.vercel.app/

- Conceived and researched a web app with 13 engineering calculators, a mission planner, and profiles of 10 real aircraft and launch vehicles.
- Wrote the specifications for each tool, including equations, inputs, and units, and used AI coding tools to write the code.
- Planned a verification page that checks results against published references (NACA Report 1135, U.S. Standard Atmosphere 1976); 26 of 36 values match within rounding.

ACADEMICS AND EXTRACURRICULARS

National Honor Society (NHS) Mar 2026 - Present
Member

Artificial Intelligence (AI) Club Sep 2025 - Present
Vice President

- Talk with other students about new AI technologies, current trends, and how AI is used in the real world.
- Build hands-on projects with my team, using AI tools to solve everyday problems.

Track and Field Team Nov 2023 - Present
Student Athlete

- Compete in the high jump, 55m hurdles, and 110m hurdles, attending daily practices and weekend track meets.
- Balance a tough class schedule with daily team practices while working hard to improve my personal records.

Robotics Club Sep 2023 - Jun 2024
Member

- Worked with a team to design, build, and test robots for VEX Robotics competitions.
- Used engineering and problem-solving skills to fix robot designs and make them perform better during challenges.

WORK EXPERIENCE

ALPHAPEP LLC, Lowell MA May 2026 - Present
Email Marketing Associate

- Designed and distributed targeted marketing emails to advertise a diverse variety of the company's products.
- Managed and executed outgoing email campaigns to ensure successful delivery of promotional materials and drive customer engagement.

LEADERSHIP AND VOLUNTEERING

BAPS Bal Mandal, Lowell MA Aug 2022 - Present
Youth Activities Coordinator *Aug 2024 - Present*
Teacher *Aug 2022 - Aug 2024*

- Helped run large youth events with over 1,000 people, focusing on teamwork and leadership.
- Managed weekly youth group meetings, organizing and leading over 70 kids and 15 peer teachers every single week.

- Planned and taught lessons for youth group meetings, using discussions and mentoring to help younger students learn cultural values and leadership skills.

BAPS Charities, Lowell MA Jun. 2017 - Present
Volunteer

- Volunteered for service projects like community cleanups, food drives, and health awareness events.
- Helped set up events, check people in, and direct crowds so everything ran smoothly.
- Encouraged others in the community to get involved and volunteer.

BAPS Akshardham, Robbinsville, NJ Jul. 2023 - Oct. 2023
Full-Time Volunteer

- Designed detailed architectural pieces using CorelDRAW and Adobe Illustrator, and created CNC machine files for a large cultural building project.
- Ran industrial CNC machines and hot wire cutters to make complex parts out of wood, PVC, and styrofoam.
- Worked with senior designers and other volunteers to make sure designs were accurate and finished on time without wasting materials.